New solvers for variational inequalities: a sample

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outline

1. active-set Newton methods
2. multilevel solvers
3. adaptive mesh refinement
4. gloss: higher-order elements, latent-variable methods

- I welcome questions at any time!
- they are a *good* use of time

recall: variational inequalities

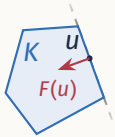
definition

For a Banach space X, a closed and convex admissible set $K \subset X$, and $F : K \to X'$ continuous, find $u \in K$ so that

$$\boxed{\langle F(u), v - u \rangle \geq 0} \qquad \text{for all } v \in K.$$

This is the variational inequality (VI) for the data K and F.

- K given via inequalities and/or equalities
  - libraries may restrict to box constraints (bounds on each dof)
- $\langle \cdot, \cdot \rangle$ denotes Hilbert space inner product or dual pairing
- this is a VI of the first kind in some references
  - second kind: $\langle F(u), v - u \rangle + \Phi(v) - \Phi(u) \geq 0 \; \forall v$, for semi-smooth $\Phi : K \to \mathbb{R}$
- well-posedness: often nontrivial

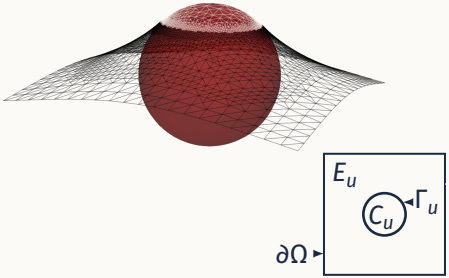


recall: classical obstacle problem as a VI

classical obstacle problem

for $\Omega \subset \mathbb{R}^d$, $X = H^1(\Omega)$, and $\psi \in X$, define $K = \{v \in X : v \geq \psi \text{ and } v|_{\partial\Omega} = g\}$, and seek $u \in K$ such that

$$\int_\Omega \nabla u \cdot \nabla(v - u) \geq \int_\Omega f(v - u) \quad \forall v \in K$$



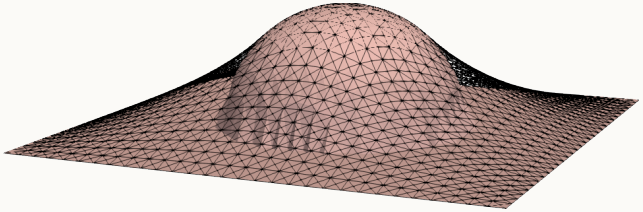
definition

a solution u defines subsets of Ω:

inactive set	$E_u = \{x : u(x) > \psi(x)\}$
active (contact) set	$C_u = \Omega \setminus E_u$
free boundary	$\Gamma_u = \Omega \cap \partial E_u$

recall: FE approximation, a finite-dimensional VI

- assume $X_h = P_1$ = continuous & piecewise-linear (I'll return to Ciarlet's gauntlet at end)
- define $\psi_h = I_h\psi$, the nodal interpolant, and $K_h = \{v_h \in X_h : v_h \geq \psi_h \text{ and } v_h|_{\partial\Omega} = g\}$
- $K_h \subset X_h \subset X = H^1(\Omega)$, but generally $K_h \not\subset K$
  - use "conforming" with care!



FE approximation of classical obstacle problem

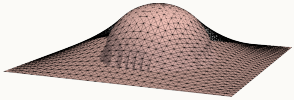
seek $u_h \in K_h$ so that $\displaystyle\int_\Omega \nabla u_h \cdot \nabla(v_h - u_h) \geq \int_\Omega f(v_h - u_h) \quad \forall v_h \in K_h$

active-set method demo: refine and count Newton iterations

```python
from firedrake import *

gridseq = False

mesh0 = RectangleMesh(10, 10, 2, 2, -2, -2, diagonal="crossed")
hierarchy = MeshHierarchy(mesh0, 5)
uh = None
for mesh in hierarchy:
    x, y = SpatialCoordinate(mesh)
    r2 = x ** 2 + y ** 2
    V = FunctionSpace(mesh, "P", 1)
    psih = Function(V).interpolate(conditional(r2 < 1, sqrt(1 - r2), 0))
    uold = uh
    uh, vh = Function(V), TestFunction(V)
    if gridseq and uold is not None:
        prolong(uold, uh)
    F = inner(grad(uh), grad(vh)) * dx
    bcs = DirichletBC(V, Constant(0.0), "on_boundary")
    prob = NonlinearVariationalProblem(F, uh, bcs)
    sp = {"snes_type": "vinewtonrsls"}  # an active-set method
    solver = NonlinearVariationalSolver(prob, solver_parameters=sp)
    INF = Function(V).interpolate(Constant(PETSc.INFINITY))
    solver.solve(bounds=(psih, INF))
    m = int(round(sqrt(mesh.num_cells() / 4)))
    q = solver.snes.getIterationNumber()
    print(f"{m:4d} x {m:4d}: {q} Newton iterations")
    #VTKFile(f"result{m}.pvd").write(uh, psih)
```



mesh	Newton its.
10 × 10	4
20 × 20	5
40 × 40	8
80 × 80	13
160 × 160	24
320 × 320	46

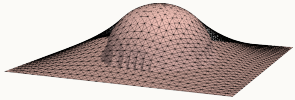
- iterations double at each refinement

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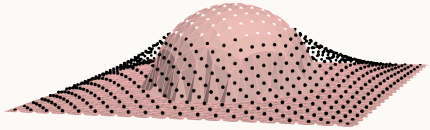
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- ~~iterations double at each refinement~~
- fixed by grid sequencing

language: residual = multipliers = dual variables

- for $u_h \in X_h = P_1$, let $\mathbf{u} = (u_i) \in \mathbb{R}^N$ denote the nodal values
- stiffness matrix A and load vector $\mathbf{b}$
- the residual is $\mathbf{F}(\mathbf{u}) = A\mathbf{u} - \mathbf{b} \in \mathbb{R}^N$
  - in classical obstacle problem, the residual $\mathbf{F}$ is a linear or affine map
- in optimization, $\lambda_i = \mathbf{F}(\mathbf{u})_i$ are the Lagrange multipliers or dual variables

complementarity problem solvers



- residual values $\lambda_i = \mathbf{F}(\mathbf{u})_i$ are the dual variables

complementarity problem (CP)

for $i \in \{1, \dots, N\}$,

$$u_i - \psi_i \geq 0, \qquad \lambda_i \geq 0, \qquad (u_i - \psi_i)\lambda_i = 0$$

- solving this CP in $\mathbb{R}^N$ is the goal in a FE VI problem
  - not "$\mathbf{F}(\mathbf{u}) = 0$"
- at convergence, in exact arithmetic, generally $\boldsymbol{\lambda} \neq 0$:

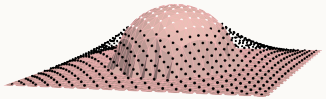
$$\lambda_i > 0 \implies u_i = \psi_i \qquad (x_i \text{ in active set } C_{u_h}, \text{ white nodes})$$

- straightforward CP generalization exists for bilateral bounds $\underline{\psi}_i \leq u_i \leq \overline{\psi}_i$

active-set method: basic idea

- VI problems are nonlinear, even when residual operator **F** is linear
  - so we expect to iterate
- consider an admissible iterate $\mathbf{u}^k \in \mathbb{R}^N$, so that $u_i^k \geq \psi_i$
- its active set is the index set (white dots)

$$\mathcal{A}^k = \{i \,:\, u_i^k = \psi_i\} \quad \subset \{1, \dots, N\}$$



  - to be precise, use tolerance: $\mathcal{A}^k = \{i \,:\, u_i^k - \psi_i \leq \text{tol}\}$
  - inactive set is complement $\mathcal{I}^k = \{1, \dots, N\} \setminus \mathcal{A}^k$ (black dots)
- if correct $\mathcal{A}^k$ is known then a single solve determines the correct $\mathbf{u} = \mathbf{u}^{k+1}$
  - for classical obstacle problem: $\mathbf{u}_{\mathcal{A}^k} = \boldsymbol{\psi}_{\mathcal{A}^k}$, $(A\mathbf{u} - \mathbf{b})_{\mathcal{I}^k} = 0$
  - but initially we do not know the active set!
- what we see in practice →

```
0 SNES Function norm 3.624969175519e+00
1 SNES Function norm 1.986779195391e+00
2 SNES Function norm 6.109441485347e-01
3 SNES Function norm 1.782534266924e-01
4 SNES Function norm 4.555052324072e-02
5 SNES Function norm 1.439049503152e-15
```

the active-set method in PETSc

- SNES type `vinewtonrsls` = `rsls` (Benson & Munson, 2006)
  - `rsls` = reduced-space with line search (reduced-space = active-set)
- designed for nonlinear $\mathbf{F} : \mathbb{R}^N \to \mathbb{R}^N$ and box constraints $\underline{\boldsymbol{\psi}} \le \mathbf{u} \le \overline{\boldsymbol{\psi}}$
  - $\underline{\psi}_i = -\infty$ or $\overline{\psi}_i = +\infty$ permitted, so unconstrained dofs o.k.
- principle: `rsls` reduces to `newtonls` if no constraints are active


S. Benson


T. Munson

`rsls`$(\boldsymbol{\psi}, \mathbf{u}^0)$**:** *unilateral case*

$\mathbf{u}^0 = \pi(\mathbf{u}^0) = \max\{u_i^0, \psi_i\}$ *truncate*

for $k = 0, 1, 2, \ldots$

$\mathcal{A}^k = \{i : u_i^k = \psi_i \text{ and } F_i(u^k) > 0\}$ & $\mathcal{I}^k = (\mathcal{A}^k)^c$

$J_{\mathbf{F}}(\mathbf{u}^k)_{\mathcal{I}^k,\mathcal{I}^k}\, \mathbf{s}_{\mathcal{I}^k} = -\mathbf{F}(\mathbf{u}^k)_{\mathcal{I}^k}$ *r. s. linear solve*

$\mathbf{s}_{\mathcal{A}^k} = 0$

$\mathbf{u}^{k+1} = \pi(\mathbf{u}^k + \alpha \mathbf{s})$ *truncated line search*

stop if (appropriate CP norm) < tol

RSLS = `vinewtonrsls`

Newton step in inactive dofs (reduced space), with line search over admissible states

- user's $\mathbf{F}$, $J_{\mathbf{F}}$ only see admissible iterates
- linear solver is user's choice (CG, multigrid, ...)
- no parameter used in setting $\mathcal{A}^k$

alternative view: semi-smooth Newton

- CP is equivalent to a semi-smooth equation:

$$u_i \geq \psi_i,\ \lambda_i \geq 0,\ \lambda_i(u_i - \psi_i) = 0 \quad \Longleftrightarrow \quad \lambda_i - \max\{0, \lambda_i - c\,(u_i - \psi_i)\} = 0$$

  - equivalent for any $c > 0$
  - $\max\{0, x\}$ is semi-smooth, with a well-behaved directional derivative

- idea: solve the VI by solving the primal-dual nonlinear system $\mathbf{G}(\mathbf{u}, \boldsymbol{\lambda}) = \mathbf{0}$:

$$\mathbf{G}(\mathbf{u}, \boldsymbol{\lambda}) = \begin{pmatrix} \mathbf{F}(\mathbf{u}) - \boldsymbol{\lambda} \\ \boldsymbol{\lambda} - \max\{\mathbf{0}, \boldsymbol{\lambda} - c(\mathbf{u} - \boldsymbol{\psi})\} \end{pmatrix}$$

- solve $\mathbf{G}(\mathbf{u}, \boldsymbol{\lambda}) = \mathbf{0}$ using Newton's method, naturally

primal-dual active-set is a Newton method

- restrict to linear case $\mathbf{F}(\mathbf{u}) = A\mathbf{u} - \mathbf{b}$, with symmetric A (quadratic optimization; $\mathbf{F} = \nabla f$)

pdas$(\boldsymbol{\psi}, \mathbf{u}^0, c)$**:** *unilateral case*

$\mathcal{A}^0 = \emptyset$, $\mathcal{I}^k = \{1, \dots, N\}$, $\boldsymbol{\lambda}^0 = \mathbf{0}$

for $k = 0, 1, 2, \dots$

$A_{\mathcal{I}^k \mathcal{I}^k} \mathbf{s}_{\mathcal{I}^k} = -A_{\mathcal{I}^k, \mathcal{A}^k}(\mathbf{u}^k - \boldsymbol{\psi}) - (A\mathbf{u}^k - \mathbf{b})_{\mathcal{I}^k}$ *linear solve*

$\mathbf{u}^{k+1}_{\mathcal{I}^k} = \mathbf{u}^k_{\mathcal{I}^k} + \mathbf{s}_{\mathcal{I}^k}$ & $\mathbf{u}^{k+1}_{\mathcal{A}^k} = \boldsymbol{\psi}_{\mathcal{A}^k}$

$\boldsymbol{\lambda}^{k+1}_{\mathcal{A}^k} = -(A\mathbf{u}^{k+1} - \mathbf{b})_{\mathcal{A}^k}$ & $\boldsymbol{\lambda}^{k+1}_{\mathcal{I}^k} = 0$

$\mathcal{A}^k = \{i : \lambda_i^k - c(u_i^k - \psi_i) > 0\}$ & $\mathcal{I}^k = \left(\mathcal{A}^k\right)^C$

stop if (appropriate CP norm) < tol

PDAS = primal-dual active-set

iterative quadratic optimization method which adjusts primal ($\mathbf{u}$) and dual ($\boldsymbol{\lambda}$) dofs via estimated active set

theorem (Hintermüller, Ito & Kunisch 2002)

- *PDAS = (semismooth Newton). Thus converges locally superlinearly.*
- *If A is M-matrix then converges globally, and iterates $k > 1$ are admissible.*

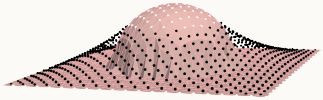

M. Hintermüller


K. Ito


K. Kunisch

clarity on iterations

- iteration counts for RSLS and PDAS have been confusing
- recent results provide clarity



theorem (Papadopoulos, Farrell, & Surowiec 2021, App. A)

On quadratic optimization problems, where **F** *is linear, PETSc's RSLS is same as PDAS, but a half-step out of sync.*

theorem ("peeling" theorem; Papadopoulos & Hintermüller 2026)

For PDAS on a quadratic opt. problem, if $i \in \mathcal{A}^k$*, and if its neighbors are in* $\mathcal{A}^k$*, then* $i \in \mathcal{A}^{k+1}$*. A node can only de-activate if has an inactive-set neighbor.*

- thus obstacle problem iterations grow exponentially
- but Signorini problem iterations only grow mildly



I. Papadopoulos



P. Farrell



T. Surowiec



M. Hintermüller

outline

1. active-set Newton methods
2. multilevel solvers
3. adaptive mesh refinement
4. gloss: higher-order elements, latent-variable methods

grid-sequencing can fix mesh-dependent iterations

- recall iterations in the Firedrake demo →
  - classical obstacle problem
  - using RSLS

mesh	RSLS only	w grid seq
10 × 10	4	4
20 × 20	5	1
40 × 40	8	2
80 × 80	13	1
160 × 160	24	2
320 × 320	46	2

- active-set method alone: Newton iterations grow fast with refinement
- with grid sequencing: mesh-independent iteration counts
  - build `MeshHierarchy()`
  - prolong a coarse solution to the next-finer grid, as initial iterate
- note that grid-sequencing is a precursor to multigrid, but it is not a V-cycle

notation for problems with nonlinear operators

- consider solving general VI, in which $F : X \to X'$ is nonlinear

definitions

Given a Banach space X and $F : X \to X'$ smooth.

PDE(F): Find $u \in X$ s.t. $\langle F(u), v\rangle = 0 \; \forall v \in X$.

VI(K,F): For $K \subset X$ closed & convex, find $u \in K$ s.t. $\langle F(u), v - u\rangle \geq 0 \; \forall v \in K$.

- PDE(F) = VI(X,F)
- functions $F : K \to X'$ are, in fact, permitted in VI(K,F)
- well-posedness, regularity are nontrivial ... next slides aspirational in general

goal: optimal solvers for VIs

- for well-behaved PDE(F) & VI(K,F), here is a sensible performance goal for FE solvers

definition

For a family of discrete problems $\{P_N\}$, where P_N has N dofs, a solver is optimal if $O(N)$ arithmetic (or time) suffices to solve P_N.

  - as $N \to \infty$, of course
  - "solve" means to discretization accuracy
  - constant in $O(N)$ definitely matters in practice
- this definition only makes sense for fixed element degree (P_1 in examples)
  - spectral and hp-FE "use dofs better" under p-refinement

recall: what's possible for PDE(F)?

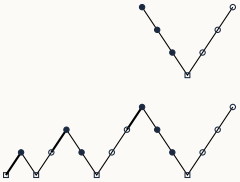
- geometric multigrid can be an optimal PDE(F) solver
  - assuming a well-behaved nonlinear elliptic F
- here is the Newton-multigrid strategy
  - principle: V-cycles are excellent linear preconditioners *e.g. (B, 2021)*



solver	optimal
Newton $\times \ell$ LU	no
Newton $\times \ell$ Krylov $\times k$ V-cycle	yes

recall: what's possible for PDE(F)?

- geometric multigrid can be an optimal PDE(F) solver
  - assuming a well-behaved nonlinear elliptic F
- here is the Newton-multigrid strategy
  - principle: V-cycles are excellent linear preconditioners *e.g. (B, 2021)*



solver	optimal
Newton $\times \ell$ → LU	no
Newton $\times \ell$ → Krylov $\times k$ → V-cycle	~~yes~~ $O(N \log N)$
grid seq → Newton $\times \ell$ → Krylov $\times k$ → V-cycle	yes $O(N)$

- adding outer grid-sequencing creates an FMG cycle

what's possible from Newton-multigrid for VI(K,F)?

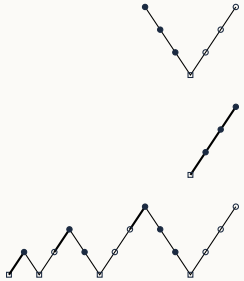
- an active-set method needs grid-sequencing to give mii
  - mii = mesh-independent (Newton) iterations
- solving the Newton-step problems by linear V-cycles is therefore not enough



solver	mii	optimal
RSLS $\times\ell$ LU	no	no
RSLS $\times\ell$ Krylov $\times k$ V-cycle	no	no

what's possible from Newton-multigrid for VI(K,F)?

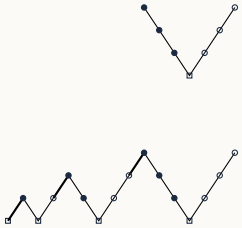
- an active-set method needs grid-sequencing to give mii
  - mii = mesh-independent (Newton) iterations
- solving the Newton-step problems by linear V-cycles is therefore not enough
- "?" means good general evidence, but no proof



solver	mii	optimal
RSLS $\times\ell$ → LU	no	no
RSLS $\times\ell$ → Krylov $\times k$ → V-cycle	no	no
grid seq → RSLS $\times\ell$ → LU	yes?	no
grid seq → RSLS $\times\ell$ → Krylov $\times k$ → V-cycle	yes?	yes?

different strategy for VI(K,F): V-cycles on the outside

- Newton-multigrid depends on grid-sequencing to find finest active set $\mathcal{A}$
  - grid-sequencing is outer loop; only FMG cycles are any good
- FASCD strategy puts RSLS inside a V-cycle (explained on next slides)
  - active-set Newton method applied as a smoother on each level



solver	optimal
FASCD V-cycle $\times\ell$	~~yes~~ $O(N \log N)$?
smoother: RSLS $\times 1$	
Krylov $\times 3$	
coarsest: RSLS $\times k$	
LU	
grid seq	yes? $O(N)$?
FASCD V-cycle $\times\ell$	
smoother: RSLS $\times 1$	
Krylov $\times 3$	
coarsest: RSLS $\times k$	
LU	

FAS multigrid and constraint decomposition

- FASCD combines 2 major ideas:

FAS: full-approximation scheme V-cycle multigrid for PDE(F) (Brandt, 1977)

At each level, restrict both residual and solution to generate a nonlinear coarse-mesh problem for the coarse correction.


A. Brandt

- advantage: no global linearization

CD: constraint decomposition for VI(K,F) (Tai, 2003)

Decompose the convex constraint set K over the mesh hierarchy: $K = \sum_{i=0}^{J} K_i$.

Compute corrections from each subproblem VI(K_i,F).

Combine them multiplicatively or additively.


X.-C. Tai

- advantage: general convergence concept/analysis
- impractical: subproblems evaluate F on finest level

FASCD = monotone multigrid + FAS + CD

- FASCD uses more ideas

- Kornhuber's (1994) monotone multigrid
- define a defect constraint set on the finest level J:
  - given admissible iterate $w \in K_h$, its defect constraint is $\chi^J = \psi_h - w \le 0$
  - let $\mathcal{U}^J = \{y : y \ge \chi^J\}$, so $0 \in \mathcal{U}^J$
  - observe: $v \in K_h \iff v = w + z$ for $z \in \mathcal{U}^J$
- also: monotone restrictions $R^{\oplus}, R^{\ominus}$


R. Kornhuber

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- also: monotone restrictions $R^{\oplus}, R^{\ominus}$


R. Kornhuber

- FASCD (B & Farrell, 2024) combines everything:
  - defect constraints are computed at start of V-cycle
  - monotone restrictions generate defect constraints on each level
  - down-smoothing in smaller sets $\mathcal{D}^j$, up-smoothing in larger sets $\mathcal{U}^j$
  - coarse-corrections solve FAS formula
  - $O(N)$ smoother: fixed RSLS iterations, fixed Krylov iterations
  - coarsest solve: RSLS+LU to convergence
  - box constraints & FMG cycles implemented


P. Farrell

FASCD

FASCD V-cycle

fascd_vcycle$(J, \ell^J, \underline{\gamma}^J, \overline{\gamma}^J; w^J)$**:**

$\underline{\chi}^J, \overline{\chi}^J = \underline{\gamma}^J - w^J, \overline{\gamma}^J - w^J$

for $j = J$ **downto** $j = 1$

$\underline{\chi}^{j-1}, \overline{\chi}^{j-1} = R^{\circledast}\underline{\chi}^j, R^{\ominus}\overline{\chi}^j$

$\underline{\phi}^j, \overline{\phi}^j = \underline{\chi}^j - P\underline{\chi}^{j-1}, \overline{\chi}^j - P\overline{\chi}^{j-1}$

$y^j = 0$

$\text{SMOOTH}^{\text{down}}(\ell^j, \underline{\phi}^j, \overline{\phi}^j, w^j; y^j)$ *VI problem in* $\mathcal{D}^j$

$w^{j-1} = R^{\bullet}(w^j + y^j)$

$\ell^{j-1} = f^{j-1}(w^{j-1}) + R\left(\ell^j - f^j(w^j + y^j)\right)$

$z^0 = 0$

$\text{SOLVE}(\ell^0, \underline{\chi}^0, \overline{\chi}^0, w^0; z^0)$ *VI problem in* $\mathcal{U}^0$

for $j = 1$ **to** $j = J$

$z^j = y^j + Pz^{j-1}$

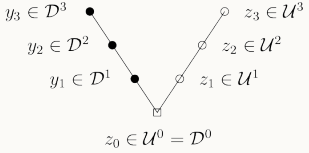
$\text{SMOOTH}^{\text{up}}(\ell^j, \underline{\chi}^j, \overline{\chi}^j, w^j; z^j)$ *VI problem in* $\mathcal{U}^j$

return $w^J + z^J$

construct defect constraints

apply FAS formulas

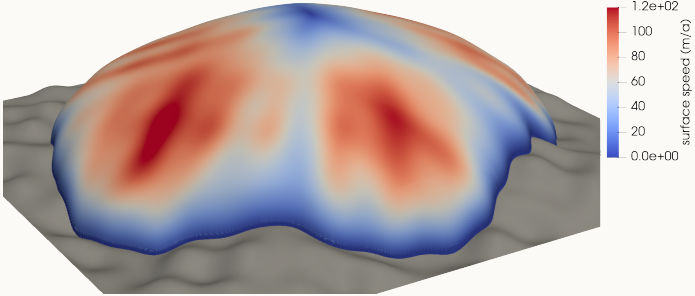
$J = 3$ case:



- nodes can activate on every level, via RSLS smoothers
- subject to peeling theorem, but on every level!

example FASCD result: a steady-state glacier

- shallow ice approximation: F is nonlinear, highly-degenerate, non-symmetric
  - VI(K,F) models steady-state glacier, directly from bed topography & climate
  - B & Farrell (2024), Example 8.4
- FASCD FMG with Q_1 elements
- $N = 4 \times 10^8$ dofs, 1024 processes



  - deep V-cycles ($J = 11$)
  - under 3 minutes wall clock
  - 88 m resolution uniformly over Greenland-sized ice sheet

outline

1. active-set Newton methods
2. multilevel solvers
3. adaptive mesh refinement
4. gloss: higher-order elements, latent-variable methods

a posteriori error estimates and AMR

- ∃ large literature of *a posteriori* estimators for classical obstacle problem
  - results which have not to have made it into much open software?
- reasonable goal: problem-robust AMR for VIs, in Firedrake of course

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- recall adaptive mesh refinement (AMR) loop:

$$\text{solve} \to \text{estimate} \to \text{mark} \to \text{refine} \to \text{solve} \to \dots$$

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- for example, consider canonical PDE estimator

Babuška & Rheinboldt (1978), for Poisson equation $-\nabla^2 u = f$

Let $\sigma_h = f + \nabla^2 u_h$. Compute $\eta_T^2 = h_T^2 \int_T |\sigma_h|^2 + \frac{h_T}{2} \int_{\partial T} [\![\nabla u_h \cdot n]\!]^2$.

For $\eta^2 = \sum_T \eta_T^2$, there are positive constants so that $c\,\eta \le |u - u_h|_{H^1} \le C\,\eta$

  - lower bound: η is efficient upper bound: η is reliable

a posteriori estimators for VIs: different character

- for VI(K,F) consider the exact residual $\lambda = F(u) \in X'$
  - it is generally a measure supported in the active set
- expectation: the map from problem data to the solution pair (u, λ) is one-to-one
  - for operators $F(u) = Lu - f$, changing f inside active set may not change u at all

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Veeser (2001), for classical obstacle problem

concept: For VIs, estimate the error *a posteriori* using the pair (u, λ).
theorem. Reliable and efficient energy-norm estimators for $(u - u_h, \lambda - \lambda_h)$.


A. Veeser


R. Nochetto

Nochetto, Siebert & Veeser (2003), for classical obstacle problem

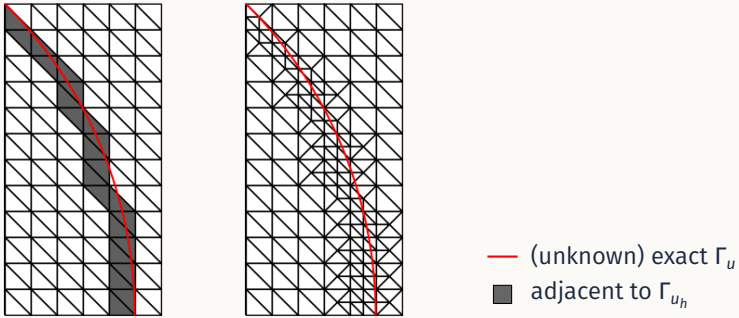
theorem. Computable pointwise estimator E_h so that
$\max\left\{\|u - u_h\|_{L^\infty},\ \|\lambda - \tilde{\lambda}_h\|_{W^{-2,\infty}}\right\} \le E_h$. Each term of E_h is also locally efficient.


K. Siebert

- E_h is complicated & expensive, even for easy classical obstacle problem

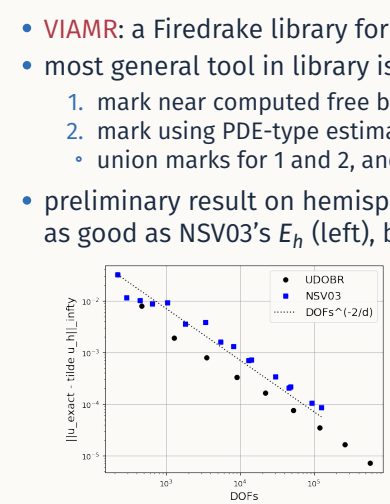
most-obvious AMR idea for VIs

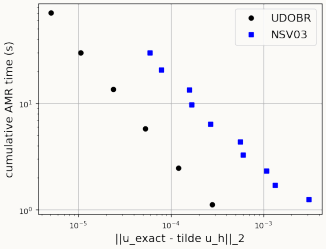
- often the free boundary location is a VI model goal, so
- mark & refine elements which are adjacent to the computed free boundary Γ_{u_h}
  - this is cheap for any VI(K,F)



library for AMR on VIs

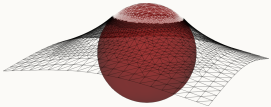
- WIP with Stefano Fochesatto (MS UAF, now at CU Boulder)
- VIAMR: a Firedrake library for easy exploration/comparison
- most general tool in library is “UDOBR”:
  1. mark near computed free boundary Γ_{u_h} (unstructured dilation)
  2. mark using PDE-type estimator in computed inactive set E_{u_h}
     - union marks for 1 and 2, and refine
- preliminary result on hemisphere classical obstacle problem: as good as NSV03’s E_h (left), but much cheaper (right)






S. Fochesatto

VIAMR



outline

1. active-set Newton methods
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3. adaptive mesh refinement
4. gloss: higher-order elements, latent-variable methods

high-order elements for VIs

- recall from last week

low-order gauntlet for VIs (Ciarlet 2002, p 289)

[Because of the low regularity in the obstacle problem], *finite elements "of low degree" are sufficient for all practical purposes, a fact amply confirmed by numerical evidence.*

- on the next two slides, ideas for higher-order FE

high-order elements in a bounded-solution PDE

- there are PDEs in which the exact solution satisfies $u(x) \in [a, b]$ for all x
  - advection-diffusion problems, for example
- if we apply ordinary P_k FE with $k \geq 2$ then u_h may not satisfy the same bounds
  - because P_2 lacks nodal monotonicity

high-order elements in a bounded-solution PDE

- there are PDEs in which the exact solution satisfies $u(x) \in [a, b]$ for all x
  - advection-diffusion problems, for example
- if we apply ordinary P_k FE with $k \geq 2$ then u_h may not satisfy the same bounds
  - because P_2 lacks nodal monotonicity

- one can enforce the bounds by having u_h solve a VI, in the right basis
- recall Bernstein basis for P_k on $x \in [0, 1]$: $B_i^k(x) = \binom{k}{i} x^i (1 - x)^{k-i}, \quad i = 0, \dots, k$
  - nonnegative partition of unity
  - (coefficients) $\geq 0 \implies p \geq 0$ (converse false)

high-order bounds-satisfying PDE solutions (Kirby & Shapero, 2024)

Using the Bernstein basis, apply VI solver with bounds on coefficients.

theorem. Convergence rate is the same as P_k, but now $u_h(x) \in [a, b]$.

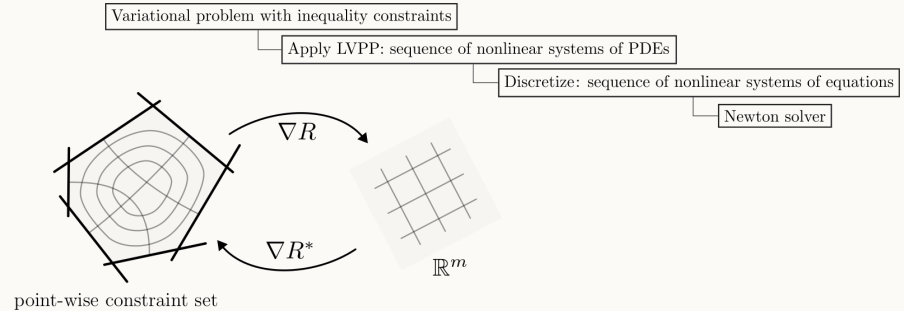

R. Kirby


D. Shapero

latent variable proximal point (LVPP)

Dokken, Farrell, Keith, Papadopoulos & Surowiec, 2025

Use a Legendre function R to map constrained point values $u(x)$ to $\mathbb{R}^m$; $m = 1$ for obstacle. Induces latent variable in new vector space: $\phi = \nabla R(u) \in W$. Enforce latent equation as: $\nabla R^*(\phi) = u \in K \subset X$. Convert VI into proximal point sequence of nonlinear PDEs. Discretize, and solve by Newton's method.




J. Dokken


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B. Keith


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T. Surowiec

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P. Farrell


B. Keith


I. Papadopoulos


T. Surowiec

- as formulas, for $\alpha_k > 0$ so that $\sum_{k=1}^{m} \alpha_k \to \infty$,

$$\langle F(u), v - u \rangle \geq 0 \quad \forall v \in K \quad \Longleftrightarrow \quad \begin{aligned} \alpha_k \langle F(u), v \rangle + \left\langle \phi^k - \phi^{k-1}, v \right\rangle &= 0 \quad \forall v \in X \\ \left\langle u^k - \nabla R^*(\phi^k), w \right\rangle &= 0 \quad \forall w \in W \end{aligned}$$

- the constraint set K does not need a discretization
- mesh-independent iterations observed
- general constraints beyond box bounds (e.g. gradient constraints)
- wide applicability (Signorini contact, fracture, plasticity, ...)

summary: my take on numerics for VI(K,F)

- active-set methods (e.g. RSLS) suffer iteration growth under refinement
  - on obstacle problems, peel away one layer per iteration
  - grid-sequencing recovers mesh-independence
- multilevel strategies:
  - seek $O(N)$ optimal solver performance
  - Newton-multigrid V-cycles unfortunately have active-set method in outer loop
    - grid sequencing still needed for mesh independence
  - FASCD V-cycles put active-set method on every level
- AMR: rigorous estimators vs heuristic refinement near free boundary
- higher-order ideas: Bernstein polynomials, latent-variable proximal point

thank you for listening!

questions?

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